Integrated Software Project

Assignment 3

Project Documentation and UML Design

Assessment information	Details
Module code	ISJ107V
Assessment weighting	20% of the final module mark
Total marks	50 marks
Due date	30 September 2026
Assessment type	Individual
Submission	Through EC

Instructions

- Answer all five questions in the order presented and retain the question number.
- Document the same approved project and emerging technology used in Assignment 2, unless a change was approved by the lecturer.
- Use a recognised UML tool. Diagrams must be readable, labelled, internally consistent and aligned with the implemented system.
- Support design and testing claims with concise evidence such as tables, screenshots, repository references and test results.
- Submit through EC.

Question 1 — Total Marks [6]

Establish the final project baseline and explain how the documented system relates to the approved proposal.

1.1 Provide the project title, system purpose, intended users, selected emerging technology and a concise executive summary of the implemented solution. (2)

1.2 Restate the final aim, measurable objectives and scope. Identify and justify any approved changes made since Assignment 2. (2)

1.3 Provide document control information: version number, date, author, revision history, glossary and referenced project artefacts. (2)

Question 2 — Total Marks [8]

Show that the final design and implementation address the approved system requirements.

2.1 Present the final prioritised functional and non-functional requirements, each with a unique identifier and measurable acceptance criterion. (3)

2.2 Create a requirements traceability matrix mapping each requirement to the relevant UML element, implementation module, test case and completion status. (3)

2.3 Explain how the implemented system satisfies the three emerging technology-specific requirements defined in Assignment 2. (2)

Recommended traceability format:

Requirement	UML reference	Module	Test case	Status
FR-01	UC-01	Authentication service	TC-01	Pass / Partial / Not met
NFR-01	Deployment model	API layer	PT-01	Pass / Partial / Not met

Question 3 — Total Marks [16]

Present the final design as implemented. Use consistent naming across all diagrams and explain important design decisions.

3.1 Provide a high-level system architecture diagram showing users, interfaces, services, data stores, external APIs and the emerging-technology component. (2)

3.2 Update the use-case diagram and provide one detailed use-case description for a core function, including normal and alternative flows. (3)

3.3 Update the class diagram to show the implemented domain classes, attributes, operations, associations and multiplicities. (3)

3.4 Provide a sequence diagram for one core transaction, including interface, control, data and emerging-technology interactions. (3)

3.5 Provide an activity diagram for one end-to-end workflow, including decisions, exceptions and completion states. (2)

3.6 Provide a component diagram and a deployment diagram showing software dependencies and the physical or cloud execution environment. (3)

Question 4 — Total Marks [10]

Document the implemented solution so that another developer could understand, operate and maintain it.

4.1 Describe the technology stack, project structure, major modules and the responsibility of each module. Include selected interface screenshots with captions. (3)

4.2 Document the database or dataset, including an entity-relationship diagram or schema, data dictionary, validation rules, privacy controls and backup approach. (3)

4.3 Document the main API or integration interface. Show endpoint or method, input, output, authentication, validation and error handling. (2)

4.4 Provide concise installation, configuration and deployment instructions, including minimum hardware and software requirements. (2)

Question 5 — Total Marks [10]

Demonstrate that the system was evaluated against its requirements and can be used responsibly.

5.1 Present a test summary covering unit, integration, system and user acceptance testing. Report the test case, expected result, actual result, status and evidence for representative tests. (4)

5.2 Evaluate at least two non-functional requirements and one emerging-technology metric using appropriate quantitative results. (2)

5.3 Provide a concise user guide for the main workflow, supported by labelled screenshots and troubleshooting guidance. (2)

5.4 Discuss current limitations, unresolved risks, ethical or security concerns, maintenance needs and two realistic future improvements. (2)